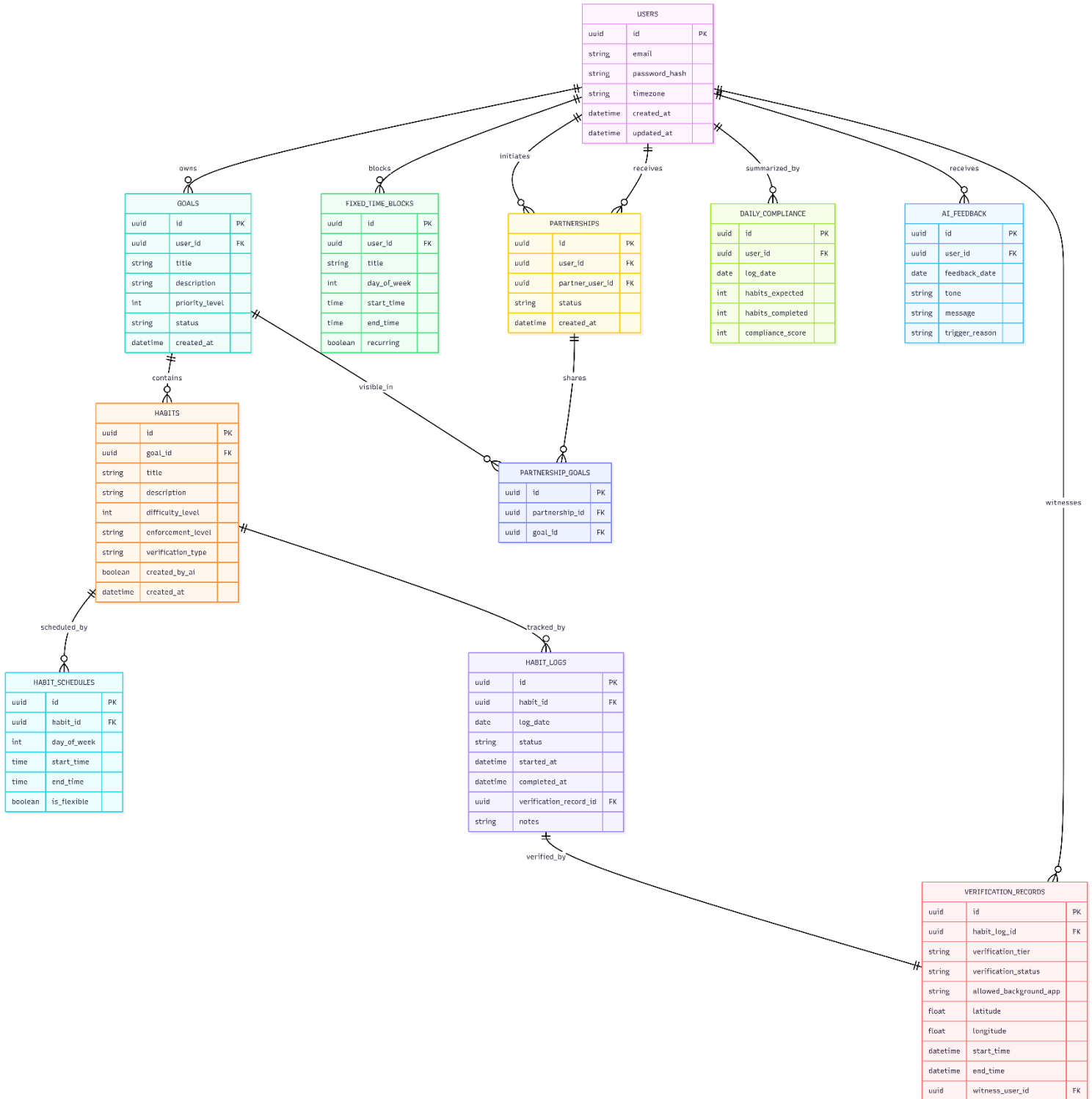


Relational Schema Justification

Accountability App Database design



1. Overview of the Relational Model

The Accountability App is designed as a **behavioral accountability system**, not a passive habit tracker. Consequently, the database schema prioritizes:

- Explicit ownership of behavioral data
- Time-bound and verifiable actions
- Social exposure as a form of enforcement
- Auditability of AI decisions and verification outcomes

A **relational database model** was selected to ensure:

- Referential integrity across user actions
- Deterministic enforcement of constraints
- Clear separation between intent, action, verification, and evaluation

All relations are normalized to **Third Normal Form (3NF)**, with deliberate storage of derived data only where performance or explainability requires it.

2. Core Entity Groups

The schema is logically divided into **five functional domains**:

1. User & Identity
2. Goal and Habit Definition
3. Time Constraints & Scheduling
4. Execution, Verification, and Logging
5. Social Accountability & Evaluation

Each domain is discussed below.

3. User & Identity Layer

USERS

The USERS relation is the **root entity** of the system.

Purpose:

- Establishes ownership of all behavioral data
- Anchors time-based enforcement through timezone awareness
- Enables bilateral social accountability relationships

Every other relation either directly or indirectly depends on USERS, ensuring that no behavioral data exists without an accountable owner.

This design choice supports:

- Multi-user scalability
- Strict data isolation
- Transparent attribution of success and failure

4. Goal and Habit Definition Layer

GOALS

The GOALS relation models **high-level user intent**.

Goals are:

- Human-defined
- Limited in number during onboarding (enforced at the application layer)
- Persistent across time

By separating goals from habits, the system avoids vague success metrics such as “goal completed” and instead focuses on **behavioral execution**.

HABITS

The HABITS relation represents **atomic, actionable behaviors** derived from goals.

Key characteristics:

- Habits belong to exactly one goal
- Habits define *what* must be done
- Habits specify enforcement and verification expectations

The `verification_type` attribute is critical: it determines how strict verification will be applied (focus session, location, social witness, etc.).

This separation allows:

- AI-generated habit restructuring
- Different verification standards per habit
- Granular accountability instead of abstract progress

5. Time Constraints & Scheduling Layer

HABIT_SCHEDULES

The HABIT_SCHEDULES relation defines **when habits must occur**.

Rationale:

- Time is treated as a first-class constraint
- Schedules are separated from habit definitions to allow rescheduling
- AI can optimize schedules without altering habit identity

This enables the system to enforce:

- Time windows
- Missed habit consequences
- Irreversible daily outcomes

FIXED_TIME_BLOCKS

The `FIXED_TIME_BLOCKS` relation stores **non-negotiable time constraints** such as school, sleep, work, or religious commitments.

Purpose:

- Prevents unrealistic AI scheduling
- Models real-world impossibility
- Improves ethical defensibility of strict enforcement

These blocks are global per user and apply across all goals and habits.

6. Execution, Verification, and Logging Layer

HABIT_LOGS

The `HABIT_LOGS` relation is the **ground truth of behavior**.

Each log represents:

- One habit
- One calendar day
- One outcome (completed, failed, missed)

This table is intentionally immutable after closure of a habit window. Retroactive edits are not permitted, mirroring real-world consequences.

The separation of logs from verification ensures that:

- A habit can be attempted
- But completion must be **proven**, not claimed

VERIFICATION_RECORDS

The VERIFICATION_RECORDS relation is central to the system's strictness.

Rather than embedding verification logic in application code, verification outcomes are **persisted as relational data**.

This table supports multiple verification modalities:

- Focus sessions with allowed background applications
- Location-based verification (e.g., gym attendance)
- Environmental proof via before/after photos
- Social witness confirmation

Key design insight:

Verification is modeled as *evidence*, not opinion.

By storing verification tier, status, timestamps, and witnesses, the system becomes:

- Auditable
- Explainable
- Resistant to user self-negotiation

7. Social Accountability Layer

PARTNERSHIPS

The PARTNERSHIPS relation models **mutual accountability agreements** between users.

Important characteristics:

- Explicit consent required
- Bilateral relationship
- Can be activated, pending, or blocked

This prevents:

- Unilateral spying
- Forced exposure
- Uncontrolled social graphs

PARTNERSHIP_GOALS

The PARTNERSHIP_GOALS relation defines **scope of visibility**.

Rather than exposing all user activity, social accountability is:

- Goal-specific
- Intentional
- Limited

This ensures that social pressure is:

- Targeted
- Relevant
- Psychologically effective without being overwhelming

8. Evaluation and Feedback Layer

DAILY_COMPLIANCE

The DAILY_COMPLIANCE relation stores **aggregated behavioral performance**.

Although derived from habit logs, it is persisted to:

- Improve performance
- Enable fast dashboard rendering
- Serve as a clean input for AI feedback generation

This is a justified denormalization commonly accepted in applied systems.

AI_FEEDBACK

The AI_FEEDBACK relation stores **system-generated accountability messages**.

Crucially:

- AI does not mutate core data
- AI produces observable outputs
- Every feedback message has a trigger reason

This design supports:

- Explainable AI behavior
- Post-hoc analysis
- Ethical transparency

9. Normalization and Integrity Guarantees

The schema satisfies **Third Normal Form (3NF)**:

- No repeating groups
- All non-key attributes depend on the full primary key
- No transitive dependencies

Referential integrity is enforced via foreign keys, ensuring:

- No orphaned records
- No unverifiable claims
- No ambiguous ownership

10. Academic and Ethical Justification

This schema is deliberately strict but ethically defensible because:

- Participation is voluntary
- Verification levels are explicit
- Social exposure is opt-in
- Rules are predictable and consistent

The system does not rely on surveillance or hidden AI judgment.

Instead, it uses **constraint-based design and social consequence**, both well-established principles in behavioral science.

11. Conclusion

The proposed relational schema provides a robust, normalized, and extensible foundation for a strict AI-driven accountability system. By explicitly modeling goals, habits, verification, and social exposure as relational entities, the system ensures transparency, auditability, and behavioral effectiveness while remaining academically rigorous.

1. 關聯式模型概述（Overview of the Relational Model）

Accountability App 被設計為一個行為問責系統（behavioral accountability system），而非被動式的習慣追蹤工具。因此，其資料庫結構設計重點在於：

- 行為資料的明確歸屬權
- 具時間約束且可驗證的行為執行
- 以社交可見性作為行為約束與強化手段
- AI 決策與驗證結果的可稽核性

選擇關聯式資料庫模型的原因在於其能夠確保：

- 使用者行為資料之間的參照完整性（Referential Integrity）
- 約束規則的確定性與可預測性執行
- 在「意圖、行為、驗證、評估」之間建立清楚且可解釋的分離

所有關聯皆符合**第三正規化（Third Normal Form, 3NF）**，僅在效能或可解釋性需求下，才有意識地儲存衍生資料。

2. 核心實體分組（Core Entity Groups）

整體資料庫結構依功能劃分為五個主要領域：

1. 使用者與身分（User & Identity）
2. 目標與習慣定義（Goal and Habit Definition）
3. 時間限制與排程（Time Constraints & Scheduling）
4. 執行、驗證與紀錄（Execution, Verification, and Logging）
5. 社交問責與評估（Social Accountability & Evaluation）

以下將逐一說明各領域之設計目的與合理性。

3. 使用者與身分層（User & Identity Layer）

USERS

USERS 為整個系統的**根實體（root entity）**。

其主要功能為：

- 建立所有行為資料的歸屬權

- 透過時區資訊支援準確的時間型行為判定
- 支援雙向的社交問責關係

系統中所有其他關聯皆直接或間接依賴 **USERS**，確保任何行為資料皆具有可追溯的責任主體。

此設計支持：

- 多使用者擴展性
- 嚴格的資料隔離
- 成功與失敗結果的透明歸因

4. 目標與習慣定義層（Goal and Habit Definition Layer）

GOALS

GOALS 關聯用以表示使用者的高層次意圖（**high-level intent**）。

其特性包括：

- 由使用者自行定義
- 在初始設定階段限制數量（於應用層實施）
- 具有長期存在性

透過將目標與實際行為分離，系統避免使用模糊的「目標完成」指標，而將重心放在可執行且可驗證的行為表現上。

HABITS

HABITS 關聯代表由目標衍生出的原子化、可執行行為。

其關鍵特性為：

- 每個習慣僅隸屬於一個目標
- 明確定義「必須完成的行為內容」
- 指定該行為所需的執行強度與驗證方式

其中，`verification_type` 屬性為核心欄位，用以決定行為所需的驗證嚴格程度（如專注時段、位置驗證、社交見證等）。

此結構使系統能夠：

- 支援 AI 對習慣結構的重新生成與調整
- 為不同習慣套用不同驗證標準
- 提供細緻的行為問責，而非抽象化進度指標

5. 時間限制與排程層（Time Constraints & Scheduling Layer）

HABIT_SCHEDULES

HABIT_SCHEDULES 關聯定義習慣必須執行的時間條件。

其設計理由包括：

- 將時間視為一級約束條件
- 排程與行為定義分離，以利重排
- AI 可在不改變行為本質的情況下最佳化時間安排

此結構支援系統執行以下約束：

- 明確的時間窗口
- 錯過行為的後果判定
- 不可逆的每日結果計算

FIXED_TIME_BLOCKS

FIXED_TIME_BLOCKS 關聯用以儲存不可協商的時間限制，如上課、睡眠、工作或宗教活動。

其目的在於：

- 防止 AI 產生不切實際的排程
- 真實建模現實世界的「不可行時間」
- 提升嚴格執行機制在倫理層面的合理性

這些時間區塊以使用者為單位，並套用至其所有目標與習慣。

6. 執行、驗證與紀錄層（Execution, Verification, and Logging Layer）

HABIT_LOGS

HABIT_LOGS 為系統中行為事實的唯一真實來源（ground truth）。

每筆紀錄代表：

- 一個習慣
- 一個日曆日
- 一種結果（完成、失敗、缺席）

在習慣時間窗口結束後，此資料表即視為不可變更，禁止事後修改，以模擬現實世界中行為後果的不可逆性。

將行為紀錄與驗證結果分離，確保：

- 行為可以被嘗試
- 但完成必須被證明，而非僅由使用者宣稱

VERIFICATION_RECORDS

VERIFICATION_RECORDS 是系統嚴格性的核心所在。

系統不將驗證邏輯隱藏於應用程式內，而是將驗證結果作為可持久化的關聯資料加以保存。

此關聯支援多種驗證方式，包括：

- 允許特定背景應用程式的專注時段驗證
- 位置驗證（如健身房出席）
- 環境證明（任務前後拍照）
- 社交見證確認

核心設計理念為：

驗證被視為「證據」，而非「主觀判斷」。

透過儲存驗證層級、狀態、時間戳與見證者，系統具備：

- 可稽核性
- 可解釋性

- 抵抗使用者自我協商與合理化的能力

7. 社交問責層（Social Accountability Layer）

PARTNERSHIPS

PARTNERSHIPS 關聯用以建模使用者之間的**雙向問責協議**。

其重要特性包括：

- 明確的雙方同意
- 對等的關係結構
- 可處於啟用、待確認或封鎖狀態

此設計避免：

- 單方面監控
- 被迫曝光
- 無限制的社交關係擴張

PARTNERSHIP_GOALS

PARTNERSHIP_GOALS 關聯定義社交可見性的**範圍控制**。

系統不會公開所有使用者行為，而是：

- 以目標為單位
- 由使用者主動選擇
- 嚴格限制曝光範圍

此設計使社交壓力得以：

- 精準施加
- 保持相關性
- 在心理層面有效而不造成過度負擔

8. 評估與回饋層（Evaluation and Feedback Layer）

DAILY_COMPLIANCE

DAILY_COMPLIANCE 關聯儲存每日彙總的行為表現結果。

雖然此資料可由行為紀錄推導而得，但仍選擇持久化儲存，以：

- 提升查詢效能
- 支援即時儀表板顯示
- 作為 AI 回饋生成的乾淨輸入來源

此屬於應用系統中合理且可接受的反正規化設計。

AI_FEEDBACK

AI_FEEDBACK 關聯用以儲存系統所生成的問責訊息。

其關鍵原則包括：

- AI 不直接修改核心行為資料
- AI 僅產生可觀察、可查詢的輸出
- 每則回饋皆具備明確的觸發原因

此設計支援：

- 可解釋的 AI 行為
- 事後分析
- 倫理與學術層面的透明性

9. 正規化與完整性保證（Normalization and Integrity Guarantees）

本資料庫結構符合第三正規化（3NF）：

- 無重複群組
- 所有非鍵屬性完全依賴主鍵
- 不存在遞移相依

透過外鍵約束實施參照完整性，確保：

- 不存在孤立紀錄
- 不存在未經驗證的完成宣稱
- 不存在責任歸屬不明的資料

10. 學術與倫理正當性（Academic and Ethical Justification）

本系統雖具高度嚴格性，但在倫理層面具備正當性，因為：

- 使用者為自願參與
- 驗證層級清楚明確
- 社交曝光採取主動選擇
- 規則具一致性與可預測性

系統不依賴隱性監控或黑箱式 AI 判斷，而是採用約束式設計與社交後果機制，這些皆為行為科學中成熟且被廣泛驗證的方法。

11. 結論（Conclusion）

本關聯式資料庫結構為一個嚴格的 AI 驅動問責系統提供了穩健、正規化且具擴展性的基礎。透過將目標、習慣、驗證與社交曝光明確建模為關聯實體，系統在保持學術嚴謹性的同時，也確保了透明性、可稽核性與行為層面的實際有效性。